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MA203 Numerical Methods

Winter Semester 2026

12.03.2026

Tutorial 1

Tutorial must be completed individually and submitted via Google Classroom by 19.03.2026 at 23:59. Ensure that the name and enrollment number is clearly written on the submission. Late submissions will not be accepted.

Question 1: [**Rounding Error**]

Perform the following operations **(i)** exactly and **(ii)** using three-digit chopping arithmetic (base 10). **(iii)** Compute the relative errors of the results obtained in part **(ii)**.

a) $\frac{4}{5} + \frac{1}{3}$.

b) $\frac{4}{5} \times \frac{1}{3}$.

c) $\left(\frac{1}{3} - \frac{3}{11}\right) + \frac{3}{20}$.

d) $\left(\frac{1}{3} + \frac{3}{11}\right) - \frac{3}{20}$.

Question 2: [**Relative Error**]

Let $f \in \mathcal{C}[a, b]$ be a function whose derivative exists on (a, b). Suppose f is to be evaluated at $x_0 \in (a, b)$, but instead of computing the actual value $f(x_0)$, the approximate value $\tilde{f}(x_0)$ is given by

$$\tilde{f}(x_0) = f(x_0 + \varepsilon).$$

a) Using the Mean Value Theorem, estimate the absolute error

$$|f(x_0) - \tilde{f}(x_0)|$$

and the relative error

$$\frac{|f(x_0) - \tilde{f}(x_0)|}{|f(x_0)|},$$

assuming $f(x_0) \neq 0$.

b) If $\varepsilon = 5 \times 10^{-6}$ and $x_0 = 1$, find bounds for the absolute and relative errors for $f(x) = e^x$ and $f(x) = \sin(x)$.

Hint: **Mean Value Theorem**: If $f \in \mathcal{C}[a, b]$ and f is differentiable on (a, b), then there exists a number $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Question 3: [**Bisection Method**]

Find an approximation of $\sqrt[3]{25}$ correct to 10^{-1} using the Bisection method.

Question 4: [**Programming**]

Write a Python program implementing the Bisection method. Using a tolerance of 10^{-4}, compute a root (if it exists) of the following functions in the given intervals:

(i) $\sqrt{x} - \cos(x)$ on $[0, 1]$.

(ii) $\sin(x) + 2$ on $[2, 6]$.

(iii) $x \exp(x)$ on $[-2, 0.5]$.

Based on your results, answer the following:

a) Does the Bisection method produce a root in each case? Explain.

b) For part **(iii)**, which stopping criterion is more appropriate: absolute error or relative error? Justify your answer.